

Partial differential equations(UNIT-II)

(For I-D and I-B sections)

A **partial differential equation(or PDE)** is an equation involving two (or more) independent variables x, y and a dependent variable z and its partial derivatives such as

$$\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}, \frac{\partial^2 z}{\partial y^2}, \text{ etc.}$$

$$\text{That is } F\left(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}, \frac{\partial^2 z}{\partial y^2}, \dots\right) = 0$$

Example: 1. $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$

2. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

Order of a PDE is the order of the highest derivative appearing in it.

Standard notations:

$$\text{We denote: } \frac{\partial z}{\partial x} = p, \quad \frac{\partial z}{\partial y} = q, \quad \frac{\partial^2 z}{\partial x^2} = r, \quad \frac{\partial^2 z}{\partial x \partial y} = s \quad \text{and} \quad \frac{\partial^2 z}{\partial y^2} = t.$$

Formation of PDE:

I. Formation of PDE by eliminating arbitrary constants:

The partial differential equation can be formed by the elimination of arbitrary constants from a relation involving three or more variables.

Let $f(x, y, z, a, b) = 0$ ---- (1) be a relation involving two arbitrary constants a, b .

Differentiate (1) partially w.r.t x and y respectively, we get,

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial x} = 0 \quad \text{---- (2)} \qquad \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial y} = 0 \quad \text{---- (3)}$$

By eliminating a, b from the equations (1), (2), (3) we get an equation of the form

$F(x, y, z, p, q) = 0$, which is a PDE of first order.

Note:

- 1. If the number of arbitrary constants equal to the number of independent variables in (1), then the P.D.E obtained by elimination is of first order.*
- 2. If the number of arbitrary constants is more than the number of independent variables, then the P.D.E obtained by elimination is of 2nd or higher order.*

Problems:

Construct the partial differential equations by eliminating arbitrary constants a , b from the following equations:

1. $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

Solution: Let $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ --- (1)

Differentiating (1) partially w.r.t. x , we get $2 \frac{\partial z}{\partial x} = \frac{2x}{a^2} + 0$ or $p = \frac{x}{a^2}$
or $a^2 = \frac{x}{p}$

Differentiating (1) partially w.r.t. y , we get $2 \frac{\partial z}{\partial y} = 0 + \frac{2y}{b^2}$ or $q = \frac{y}{b^2}$
or $b^2 = \frac{y}{q}$

Substituting these values of a^2 and b^2 in (1), we get

$$2z = \frac{x^2}{\left(\frac{x}{p}\right)} + \frac{y^2}{\left(\frac{y}{q}\right)}$$

that is, $2z = px + qy$ as the desired PDE of first order.

2. $z = (x - a)^2 + (y - b)^2 + 1$

Solution: Let $z = (x - a)^2 + (y - b)^2 + 1$ --- (1)

Differentiating (1) partially w.r.t. x , we get $\frac{\partial z}{\partial x} = 2(x - a) + 0$

or $(x - a) = \frac{p}{2}$

Differentiating (1) partially w.r.t. y , we get $\frac{\partial z}{\partial y} = 2(y - b)$

or $y - b = \frac{q}{2}$

Substituting these values of $x - a$ and $y - b$ in (1), we get

$$z = \left(\frac{p}{2}\right)^2 + \left(\frac{q}{2}\right)^2 + 1 \quad \text{or} \quad z = \frac{p^2}{4} + \frac{q^2}{4} + 1$$

that is, **$4z = p^2 + q^2 + 4$** as the desired PDE of first order.

$$3. z = a \log \left\{ \frac{b(y-1)}{1-x} \right\}$$

Solution: We can rewrite, $z = a \log b + a \log(y - 1) - a \log(1 - x)$ ----- (1)

Differentiating (1) partially w.r.t. x, we get $\frac{\partial z}{\partial x} = 0 + 0 - \frac{a}{1-x} \cdot (-1)$ or $p = \frac{a}{1-x}$

$$\text{or} \quad a = p(1 - x) \quad \text{----- (2)}$$

Differentiating (1) partially w.r.t. y, we get $\frac{\partial z}{\partial y} = \frac{a}{y-1}$ or $q = \frac{a}{y-1}$

$$a = q(y - 1) \quad \text{----- (3)}$$

Eliminate **a** from the equations (2) and (3), we get

$$p(1 - x) = a = q(y - 1)$$

$$p(1 - x) = q(y - 1)$$

$$\text{Or} \quad xp + yq = p + q$$

II. Formation of PDE by eliminating arbitrary Functions:

a) Equations involving one arbitrary function:

Consider $z = f(u)$ --- (4),

where $f(u)$ is an arbitrary function of u and $u = u(x, y, z)$.

Differentiating (4) w.r.t. x and y respectively, we get

$$\frac{\partial z}{\partial x} = \frac{df}{du} \cdot \frac{\partial u}{\partial x} + \frac{df}{du} \cdot \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial x} \quad \text{--- (5)}$$

$$\frac{\partial z}{\partial y} = \frac{df}{du} \cdot \frac{\partial u}{\partial y} + \frac{df}{du} \cdot \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial y} \quad \text{--- (6)}$$

By eliminating arbitrary function from the equations (4), (5), (6), we get a PDE of first order.

b) Equations involving two or more arbitrary function:

Differentiating twice or more, the elimination process results in a PDE of 2nd or higher order.

***Note:** When n is the number of arbitrary functions, one may get several partial differential equations. But generally the one with the least order is chosen.*

Form the partial differential equations by eliminating arbitrary functions:

1. $z = f(x^2 - y^2)$

Solution: Let $z = f(x^2 - y^2)$ ----- (1)

Differentiating equation (1) partially w.r.t. x , we get $\frac{\partial z}{\partial x} = f'(x^2 - y^2) \cdot 2x$
or $p = 2x \cdot f'(x^2 - y^2)$ ---- (2)

Differentiating equation (1) partially w.r.t. y , we get $\frac{\partial z}{\partial y} = f'(x^2 - y^2) \cdot (-2y)$
or $q = -2y \cdot f'(x^2 - y^2)$ ---- (3)

To eliminate $f'(x^2 - y^2)$ from (2) and (3), Divide (2) by (3), we get

$$\frac{p}{q} = \frac{2x \cdot f'(x^2 - y^2)}{-2y \cdot f'(x^2 - y^2)} = \frac{-x}{y} \quad \text{or} \quad \frac{p}{q} = \frac{-x}{y}$$

that is, $py = -xq$ or $xq + yp = 0$ as the desired PDE.

$$2. \quad z = f(x) + e^y g(x)$$

Solution: Let $z = f(x) + e^y g(x)$ ----- (1)

Differentiating equation (1) partially w.r.t. y , we get $\frac{\partial z}{\partial y} = e^y g(x)$ ----- (2)

or $q = e^y g(x)$ ----- (3)

Differentiating equation (2) partially w.r.t. y , we get $\frac{\partial^2 z}{\partial y^2} = e^y g(x)$ ----- (4)

or $t = e^y g(x)$ ----- (5)

From the equations (3) and (5), we get $q = e^y g(x) = t$

$\therefore$ The required PDE is $q - t = 0$.

$$3. \quad z = f(x + iy) + g(x - iy)$$

$$\text{Solution: Let } z = f(x + iy) + g(x - iy) \quad \text{--- (1)}$$

$$\text{Differentiating equation (1) partially w.r.t. } x, \text{ we get } \frac{\partial z}{\partial x} = f'(x + iy) \cdot 1 + g'(x - iy) \cdot 1$$

$$\text{or} \quad p = f'(x + iy) + g'(x - iy) \quad \text{--- (2)}$$

$$\text{Differentiating equation (1) partially w.r.t. } y, \text{ we get } \frac{\partial z}{\partial y} = f'(x + iy) \cdot i + g'(x - iy) \cdot (-i)$$

$$\text{or} \quad q = i \cdot f'(x + iy) - i \cdot g'(x - iy) \quad \text{--- (3)}$$

$$\text{Differentiating equation (2) partially w.r.t. } x, \text{ we get } \frac{\partial^2 z}{\partial x^2} = f''(x + iy) \cdot 1 + g''(x - iy) \cdot 1$$

$$\text{or} \quad \frac{\partial^2 z}{\partial x^2} = f''(x + iy) + g''(x - iy) \quad \text{--- (4)}$$

$$\text{Differentiating equation (3) partially w.r.t. } y, \text{ we get } \frac{\partial^2 z}{\partial y^2} = f''(x + iy) \cdot i^2 + g''(x - iy) \cdot (-i)^2$$

$$\text{or} \quad \frac{\partial^2 z}{\partial y^2} = -f''(x + iy) - g''(x - iy) \quad \text{--- (5) } (\because i^2 = -1)$$

From the equation (4) + (5), we get

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = f''(x + iy) + g''(x - iy) - f''(x + iy) - g''(x - iy) = 0$$

$$\text{or} \quad \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \mathbf{0} \text{ as the desired PDE of 2}^{\text{nd}} \text{ order.}$$

$$4. \quad x + y + z = f(x^2 + y^2 + z^2)$$

Solution: Let $x + y + z = f(x^2 + y^2 + z^2)$ -----(1)

Differentiating equation (1) partially w.r.t x , we get $1 + 0 + \frac{\partial z}{\partial x} = f'(x^2 + y^2 + z^2) \cdot (2x + 0 + 2z \frac{\partial z}{\partial x})$
or $1 + p = 2f'(x^2 + y^2 + z^2) \cdot (x + zp)$ -----(2)

Differentiating equation (1) partially w.r.t y , we get $0 + 1 + \frac{\partial z}{\partial y} = f'(x^2 + y^2 + z^2) \cdot (0 + 2y + 2z \frac{\partial z}{\partial y})$
or $1 + q = 2f'(x^2 + y^2 + z^2) \cdot (y + zq)$ -----(3)

To eliminate $f'(x^2 + y^2 + z^2)$ from (2) and (3), Divide (2) by (3), we get

$$\frac{1+p}{1+q} = \frac{2f'(x^2+y^2+z^2) \cdot (x+zp)}{2f'(x^2+y^2+z^2) \cdot (y+zq)} = \frac{x+zp}{y+zq} \quad \text{or} \quad \frac{1+p}{1+q} = \frac{x+zp}{y+zq}$$

that is, $(1 + p)(y + zq) = (1 + q)(x + zp)$

or $y + py + zq + zpq = x + xq + zp + zpq$

or $(y - z)p - (x - z)q = x - y$ as the desired PDE.

c) Formation of PDE by eliminating arbitrary Functions F

from the equation

$$F(u, v) = 0 \quad \text{----- (7),}$$

where $u = u(x, y, z)$ and $v = v(x, y, z)$ are functions of x, y, z .

Differentiate (7) partially w. r. t. x and y and eliminate arbitrary functions. Then we will get required PDE of first order.

Form the partial differential equations by eliminating arbitrary functions:

1. $F(x^2 + y^2, z - xy) = 0$

Solution: Let $u = x^2 + y^2$ and $v = z - xy$

$$\text{Then, } F(u, v) = F(x^2 + y^2, z - xy) = 0 \quad \text{--- (1)}$$

Differentiating equation (1) partially w.r.t x , we get

$$\frac{\partial F}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial F}{\partial v} \cdot \frac{\partial v}{\partial x} = 0 \quad \text{that is,} \quad \frac{\partial F}{\partial u} \cdot (2x + 0) + \frac{\partial F}{\partial v} \cdot \left(\frac{\partial z}{\partial x} - y \right) = 0$$

$$2x \frac{\partial F}{\partial u} + (p - y) \frac{\partial F}{\partial v} = 0$$

$$2x \frac{\partial F}{\partial u} = -(p - y) \frac{\partial F}{\partial v} \quad \text{--- (2)}$$

Differentiating equation (1) partially w.r.t y, we get

$$\frac{\partial F}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial F}{\partial v} \cdot \frac{\partial v}{\partial y} = 0 \quad \text{that is,} \quad \frac{\partial F}{\partial u} \cdot (0 + 2y) + \frac{\partial F}{\partial v} \cdot \left(\frac{\partial z}{\partial y} - x \right) = 0$$

$$2y \frac{\partial F}{\partial u} + (q - x) \frac{\partial F}{\partial v} = 0$$

$$2y \frac{\partial F}{\partial u} = -(q - x) \frac{\partial F}{\partial v} \quad \text{--- (3)}$$

Dividing (2) by (3), we get

$$\frac{2x \frac{\partial F}{\partial u}}{2y \frac{\partial F}{\partial u}} = \frac{-(p-y) \frac{\partial F}{\partial v}}{-(q-x) \frac{\partial F}{\partial v}} \Rightarrow \frac{x}{y} = \frac{p-y}{q-x} \quad \text{or} \quad x(q-x) = y(p-y)$$

$\therefore xq - yp = x^2 - y^2$ as the desired first order PDE.

$$2. F(x^2 + 2yz, y^2 + 2xz) = 0$$

Solution: Let $u = x^2 + 2yz$ and $v = y^2 + 2xz$

$$\text{Then, } F(u, v) = F(x^2 + 2yz, y^2 + 2xz) = 0 \quad \text{--- (1)}$$

Differentiating equation (1) partially w.r.t x , we get

$$\frac{\partial F}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial F}{\partial v} \cdot \frac{\partial v}{\partial x} = 0 \quad \text{that is,} \quad \frac{\partial F}{\partial u} \cdot \left(2x + 2y \frac{\partial z}{\partial x}\right) + \frac{\partial F}{\partial v} \cdot \left(2z + 2x \frac{\partial z}{\partial x}\right) = 0$$

$$(x + yp) \frac{\partial F}{\partial u} + (z + xp) \frac{\partial F}{\partial v} = 0$$

$$(x + yp) \frac{\partial F}{\partial u} = -(z + xp) \frac{\partial F}{\partial v} \quad \text{--- (2)}$$

Differentiating equation (1) partially w.r.t y , we get

$$\frac{\partial F}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial F}{\partial v} \cdot \frac{\partial v}{\partial y} = 0 \quad \text{that is,} \quad \frac{\partial F}{\partial u} \cdot \left(2z + 2y \frac{\partial z}{\partial y}\right) + \frac{\partial F}{\partial v} \cdot \left(2y + 2x \frac{\partial z}{\partial y}\right) = 0$$

$$(z + yq) \frac{\partial F}{\partial u} + (y + xq) \frac{\partial F}{\partial v} = 0$$

$$(z + yq) \frac{\partial F}{\partial u} = -(y + xq) \frac{\partial F}{\partial v} \quad \text{--- (3)}$$

Dividing (2) by (3), we get

$$\frac{(x+yp)\frac{\partial F}{\partial u}}{(z+yq)\frac{\partial F}{\partial u}} = \frac{-(z+xp)\frac{\partial F}{\partial v}}{-(y+xq)\frac{\partial F}{\partial v}} \Rightarrow \frac{(x+yp)}{(z+yq)} = \frac{z+xp}{y+xq}$$

or $(x + yp)(y + xq) = (z + xp)(z + yq)$

or $xy + x^2q + y^2p + xypq = z^2 + xzp + yzq + xypq$

or $xy + x^2q + y^2p = z^2 + xzp + yzq$

or $(y^2 - xz)p + (x^2 - yz)q = z^2 - xy.$

Problems:

Obtain the PDE by eliminating arbitrary constants/functions:

1. $z = ax^2 + by^2$

2. $z = ax + by + a^2 + b^2$

3. $xyz = f(x + y + z)$

4. $F(x + y + z, x^2 + y^2 + z^2) = 0$

5. $z = f(y + 2x) + g(y - 3x) \left[\text{ans: } \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6 \frac{\partial^2 z}{\partial y^2} = 0 \right]$

Solution of a Partial differential equation:

The general form of a first order PDE is $F(x, y, z, p, q) = 0$ ----- (1) where x and y are independent variables, z is dependent variable and $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$.

A **solution** or **integral** of the PDE is the relation between the dependent and independent variables satisfying the equation.

Complete solution:

Any function $f(x, y, z, a, b) = 0$ ----- (2) involving two arbitrary constants a , b and satisfying the PDE (1) is known as complete solution or complete integral or primitive.

Ex: $z = (x + a)(y + b)$ is the complete solution of the PDE $z = pq$.

Particular solution:

A solution obtained by giving particular value to the arbitrary constants in the complete solution is called a particular solution of the PDE.

Ex: $z = (x + 2)(y + 4)$ is the particular solution of the PDE $z = pq$.

General solution:

In the complete solution if we put $b = \varphi(a)$ then we get a solution containing an arbitrary function φ , which is called a general solution of PDE (1). It represents the envelope of the family of surfaces $f(x, y, z, a, \varphi(a)) = 0$.

Method of separation of variables:

For PDE in the function of u (or z) of 2 independent variables x and y , assume that the required solution is separable,

$$\text{i.e., } u = XY \quad \text{--- (1), where } X = X(x) \text{ is a function of } x \text{ alone} \\ \text{and } Y = Y(y) \text{ is a function of } y \text{ alone.}$$

Then substitution of u from (1) and its derivatives reduces the given PDE to the form

$$f(X, X', X'', \dots) = g(Y, Y', Y'', \dots) \quad \text{--- (2) which is separable in } X \text{ and } Y.$$

Since LHS of (2) is function of x alone and RHS is function of only y (x, y are independent, therefore, (2) can only be true if each side is equal to the same constant), hence (2) must be equal to a common constant say k . Thus (2) reduces to,

$$f(X, X', X'', \dots) = k \quad \text{--- (3)}$$

$$g(Y, Y', Y'', \dots) = k \quad \text{--- (4)}$$

Thus the determination of solution of PDE reduces to the determination of solutions to two ordinary differential equations(ODEs).

Solve the following partial differential equations by the method of separation of variables:

1. $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$

Solution: Let $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$ --- (1),

assume that the required solution of (1) is of the form

$$z = XY \quad \text{where } X = X(x), \quad Y = Y(y)$$

$$\therefore \frac{\partial z}{\partial x} = Y \frac{\partial X}{\partial x} = Y \frac{dX}{dx} = YX' \quad \left(\text{Here } \frac{\partial X}{\partial x} = \frac{dX}{dx}, \text{ because } X = X(x) \text{ is a function of } x \text{ alone} \right)$$

$$\text{also, } \frac{\partial z}{\partial y} = X \frac{\partial Y}{\partial y} = X \frac{dY}{dy} = XY' \quad \text{and} \quad \frac{\partial^2 z}{\partial x^2} = YX''$$

Substitute these values in (1), we have

$$YX'' - 2YX' + XY' = 0 \quad \text{or} \quad Y(X'' - 2X') = -XY'$$

$$\text{separating the variables, we get: } \frac{X'' - 2X'}{X} = -\frac{Y'}{Y} \quad \text{--- (2)}$$

Since x, y are independent, therefore, (2) can only be true if each side is equal to the same constant, say k .

$$\therefore \frac{X'' - 2X'}{X} = k \quad \text{i.e., } X'' - 2X' = kX \quad \text{or} \quad X'' - 2X' - kX = 0 \quad \text{---- (3)}$$

$$\text{and } -\frac{Y'}{Y} = k \quad \text{i.e., } Y' = -kY \quad \text{--- (4)}$$

Here (3) are (4) are ordinary differential equations,

To solve the ordinary linear differential equation (3), the auxiliary equation is:

$$m^2 - 2m + k = 0, \quad \therefore m = 1 \pm \sqrt{1+k}$$

$\therefore$ the solution of (3) is: $\mathbf{X} = c_1 e^{(1+\sqrt{1+k})x} + c_2 e^{(1-\sqrt{1+k})x}$

Equation (4) is: $\frac{dY}{dy} = -kY$ or $\frac{dY}{Y} = -k dy$

which is variable separable differential equation,

$\therefore$ the solution of (4) is: $\log Y = -ky + c_3'$

$$\Rightarrow Y = e^{-ky+c_3'} = e^{-ky} e^{c_3'} = c_3 e^{-ky} \quad \text{where } c_3 = e^{c_3'}$$

$$\Rightarrow Y = c_3 e^{-ky}$$

$$\therefore z = \mathbf{XY} = \{c_1 e^{(1+\sqrt{1+k})x} + c_2 e^{(1-\sqrt{1+k})x}\} c_3 e^{-ky}$$

$$\text{i.e., } \mathbf{z} = \{\mathbf{d_1} e^{(1+\sqrt{1+k})x} + \mathbf{d_2} e^{(1-\sqrt{1+k})x}\} \mathbf{e^{-ky}} \quad [\text{where } d_1 = c_1 c_3 \text{ and } d_2 = c_2 c_3]$$

which is the required complete solution.

$$2. \frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u, \text{ where } u(x, 0) = 6e^{-3x}$$

Solution: Let $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$ --- (1),

assume that the required solution of (1) is of the form

$$u = XT \quad \text{where } X = X(x), \quad T = T(t)$$

$$\therefore \frac{\partial u}{\partial x} = T \frac{\partial X}{\partial x} = T \frac{dX}{dx} = TX' \quad \left(\text{Here } \frac{\partial X}{\partial x} = \frac{dX}{dx}, \text{ because } X = X(x) \text{ is a function of } x \text{ alone} \right)$$

$$\text{also } \frac{\partial u}{\partial t} = X \frac{\partial T}{\partial t} = X \frac{dT}{dt} = XT'$$

Substitute these values in (1), we have

$$TX' = 2XT' + XT = X(2T' + T)$$

$$\text{separating the variables, we get } \frac{X'}{X} = \frac{2T' + T}{T} \quad \text{--- (2)}$$

Since x, y are independent, therefore, (2) can only be true if each side is equal to the same constant, say k.

$$\therefore \frac{X'}{X} = k \quad \text{i.e., } X' = kX \quad \text{--- (3)}$$

$$\text{and } \frac{2T' + T}{T} = k \quad \text{i.e., } 2T' + T = kT \quad \text{or } 2T' = (k - 1)T \quad \text{--- (4)}$$

Here (3) and (4) are ordinary differential equations,

Equation (3) is: $\frac{dX}{dx} = kX$ or $\frac{dX}{X} = k dx$ which is variable separable,

$\therefore$ the solution is $\log X = kx + c_1'$

$$\Rightarrow X = e^{kx+c_1'} = e^{kx} e^{c_1'} = c_1 e^{kx} \quad \text{where } c_1 = e^{c_1'}$$

$$\Rightarrow X = c_1 e^{kx}$$

Equation (4) is: $2 \frac{dT}{dt} = (k-1)T$ or $\frac{2dT}{T} = (k-1)dt$ which is also variable separable,

$\therefore$ the solution is $2 \log T = (k-1)t + 2c_2'$

$$\Rightarrow T = e^{\frac{1}{2}(k-1)t+c_2'} = e^{\frac{1}{2}(k-1)t} e^{c_2'} = c_2 e^{\frac{1}{2}(k-1)t} \quad \text{where } c_2 = e^{c_2'}$$

$$\Rightarrow T = c_2 e^{\frac{1}{2}(k-1)t}$$

$$\therefore u = XT = c_1 e^{kx} c_2 e^{\frac{1}{2}(k-1)t}$$

$$\therefore u = c_3 e^{kx + \frac{1}{2}(k-1)t}$$

Given that, when $(x, t) = (x, 0)$, $u(x, 0) = 6e^{-3x} = c_3 e^{kx + \frac{1}{2}(k-1).0} = c_3 e^{kx}$

$$\Rightarrow c_3 = 6 \quad \text{and} \quad k = -3$$

$$\therefore u = 6e^{-3x-2t} \quad \text{as the required particular solution.}$$

$$3. \text{ } y^3 \frac{\partial z}{\partial x} + x^2 \frac{\partial z}{\partial y} = 0$$

Solution: Let $y^3 \frac{\partial z}{\partial x} + x^2 \frac{\partial z}{\partial y} = 0 \quad \text{--- (1),}$

assume that the required solution of (1) is of the form

$$z = XY \quad \text{where } X = X(x), \quad Y = Y(y)$$

$$\therefore \frac{\partial z}{\partial x} = Y \frac{\partial X}{\partial x} = Y \frac{dX}{dx} = YX'$$

also, $\frac{\partial z}{\partial y} = X \frac{\partial Y}{\partial y} = X \frac{dY}{dy} = XY'$

Substitute these values in (1), we have

$$y^3 YX' + x^2 XY' = 0 \quad \text{or} \quad y^3 YX' = -x^2 XY'$$

separating the variables, we get: $\frac{X'}{x^2 X} = -\frac{Y'}{y^3 Y} \quad \text{--- (2)}$

Since x, y are independent, therefore, (2) can only be true if each side is equal to the same constant, say k.

$$\therefore \frac{X'}{x^2 X} = k \quad \text{i.e., } X' = kx^2 X \quad \text{--- (3)}$$

and $-\frac{Y'}{y^3 Y} = k \quad \text{i.e., } Y' = -ky^3 Y \quad \text{--- (4)}$

Equation (3) is: $\frac{dX}{dx} = kx^2 X$ or $\frac{dX}{X} = kx^2 dx$ which is variable separable,

$\therefore$ the solution is $\log X = \frac{kx^3}{3} + c_1'$

$$\Rightarrow X = e^{\frac{kx^3}{3} + c_1'} = e^{\frac{kx^3}{3}} e^{c_1'} = c_1 e^{\frac{kx^3}{3}} \quad \text{where } c_1 = e^{c_1'}$$

$$\Rightarrow X = c_1 e^{\frac{kx^3}{3}}$$

Equation (4) is: $\frac{dY}{dy} = -ky^3 Y$ or $\frac{dY}{Y} = -ky^3 dy$ which is also variable separable,

$\therefore$ the solution is $\log Y = -\frac{ky^4}{4} + c_2'$

$$\Rightarrow Y = e^{-\frac{ky^4}{4} + c_2'} = e^{-\frac{ky^4}{4}} e^{c_2'} = c_2 e^{-\frac{ky^4}{4}} \quad \text{where } c_2 = e^{c_2'}$$

$$\Rightarrow Y = c_2 e^{-\frac{ky^4}{4}}$$

$$\therefore u = XY = c_1 e^{\frac{kx^3}{3}} c_2 e^{-\frac{ky^4}{4}}$$

$$\therefore u = c_3 e^{k\left(\frac{x^3}{3} - \frac{y^4}{4}\right)}.$$

Solve the following partial differential equations by the method of separation of variables:

1. $u_{xx} - u_{yy} = 0$

2. $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = 0$

Classification of second order PDE:

The general form of a second-order P.D.E. in the function u of the two independent variables x, y is given by

$$A(x, y) \frac{\partial^2 u}{\partial x^2} + B(x, y) \frac{\partial^2 u}{\partial x \partial y} + C(x, y) \frac{\partial^2 u}{\partial y^2} + f \left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right) = 0 \quad (1)$$

This equation is linear in second order terms. PDE (1) is said to be “linear or quasi-linear” according as f is linear or non-linear.

PDE (1) is classified as *elliptic*, *parabolic* or *hyperbolic* according as $B^2 - 4AC < 0$, $B^2 - 4AC = 0$ or $B^2 - 4AC > 0$ respectively.

Example:

Elliptic: ($B^2 - 4AC < 0$)

Laplace's equation in two dimensions

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad (2)$$

Poisson's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y) \quad (3)$$

Parabolic: ($B^2 - 4AC = 0$)

One dimensional heat-flow equation

$$a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t} \quad (4)$$

Hyperbolic: ($B^2 - 4AC > 0$)

One-dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} \quad (5)$$

Note: Go through the problems and examples which have solved in the class.